

SABIR ALI SIDDIQUE (PhD)

Lecturer, Institute of Chemistry, The Islamia University of Bahawalpur, Pakistan

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RESEARCH PROFILE

Computational chemist with experience in DFT, TD-DFT, and periodic electronic-structure calculations for molecular and materials systems. My research focuses on electronic excited states, charge-transfer phenomena, optical properties, and structure–property relationships in chemically relevant systems. I work with Gaussian, VASP, Quantum ESPRESSO, AMS, and Python-based workflows, and I increasingly integrate machine learning and automated screening into electronic-structure research. My long-term interests lie in electron transfer, excited-state phenomena, and predictive computational modeling across complex molecular environments.

RESEARCH INTERESTS

TD-DFT; excited-state electronic structure; charge-transfer and redox processes; periodic and molecular DFT; electron transfer in solvated environments; multiscale computational chemistry; machine learning for chemistry and materials.

EDUCATION

Ph.D. in Chemistry | University of The Punjab, Lahore 2019–2024

Advisor: Prof. Dr. Ejaz Ahmed & Dr. Tariq Mahmood

Dissertation: Sensing of Toxic Chemicals using DFT and Optimization of Organic Solar Cells

Award: Best Research Student (2021)

M.Phil. in Chemistry | University of The Punjab, Lahore 2016–2018

Advisor: Prof. Dr. Muhammad Arshad & Dr. Khurshid Ayub

Dissertation: Hydrogen Storage in Transition Metal-Encapsulated Nanocages

B.S. Chemistry (Hons.) | University of Education, Lahore 2011–2015

RESEARCH EXPERIENCE

Lecturer | Institute of Chemistry, The Islamia University of Bahawalpur, Pakistan Sep 2022 – Present

- Lead computational research projects in DFT, TD-DFT, and molecular/materials modeling.
- Supervise projects focused on electronic structure, excited-state properties, charge transfer, and structure–property relationships in functional chemical systems.
- Develop machine-learning-assisted workflows for predictive modeling in electronic-structure and computational chemistry projects.

Ph.D. Researcher | University of The Punjab, Lahore, Pakistan Sep 2019 – Dec 2024

- Carried out DFT and TD-DFT studies on gas sensing, optoelectronic materials, NLO systems, and photovoltaics.
- Analyzed excited states, charge-transfer behavior, electronic spectra, and frontier orbital characteristics in molecular systems.
- Published multiple first-author and co-authored papers on electronic structure, excited-state properties, and functional materials modeling.

Research Associate | Lahore University of Management Sciences (LUMS), Pakistan Feb 2020 – May 2021

- Modeled energy-related materials and photoactive systems in computational projects supported by national research funding.
- Performed first-principles calculations and post-processing for materials-property analysis.

SELECTED PUBLICATIONS

Total: 37+ peer-reviewed publications;

(900+ Citations | h-index : 16 | i10 index : 21)

1. Siddique, S. A. et al. *Discovery of pyrrole-triphenylamine based novel organic sensitizers for dye-sensitized solar cells: A first-principles study*. Materials Science in Semiconductor Processing, 2024, 174, 108173.
2. Siddique, S. A. et al. *Efficient tuning of zinc phthalocyanine-based dyes for dye-sensitized solar cells: A detailed DFT study*. RSC Advances, 2021, 11, 27570–27582.
3. Siddique, S. A. et al. *Discovery of versatile bat-shaped acceptor materials for high-performance organic solar cells – A DFT approach*. International Journal of Energy Research, 2022, 46, 13393–13408.
4. Siddique, S. A. et al. *Quantum chemical engineering of enhanced nonlinear optical responses in alkali metal-doped diazadioxacirculenes for molecular photonics*. Journal of Molecular Graphics and Modelling, 2026, 142, 109229.
5. Siddique, S. A. et al. *Quantum chemical investigation for enhanced electrochemical sensing of toxic gases by hexaazaphenH₂*. New Journal of Chemistry, 2024, 48, 18128–18142.

Full publication list available on request or via [Google Scholar](#).

SUPERVISION & TEACHING

Supervised 15+ BS and M.Phil. theses in DFT, TD-DFT, photovoltaics, catalysis, and computational materials modeling.

Taught Quantum Chemistry, Molecular Spectroscopy, and Computational Chemistry.

HONORS & AWARDS

Emerging Young Research Mentor Institute of Chemistry, The Islamia University of Bahawalpur, Pakistan	2023
Research Excellence Award University of The Punjab, Lahore	2021
Academic Excellence Scholarship University of Education, Lahore	2014
PM Laptop Merit Award University of Education, Lahore	2012

TECHNICAL SKILLS

Electronic structure and simulation: VASP, Gaussian, Quantum ESPRESSO, Materials Studio, AMS, ASE

Methods: DFT, TD-DFT, periodic DFT, molecular modeling, excited-state analysis, band-structure analysis, charge-transfer analysis

Programming and workflows: Python, Bash, C++, R

Machine Learning: scikit-learn, SchNetPack, DeePMD-kit

COLLABORATIONS, REVIEWING & OUTREACH

Collaborations: King Saud University, University of Malaya, Shandong University, University of Rome Tor Vergata, Dalian University, Xi'an Jiaotong University, South Ural State University.

Reviewer for journals including: *ACS Omega*, *Chemosphere*, *RSC Advances*, and *Solar Energy* (50+ manuscripts)

STEM workshops for high school students in Pakistan, hands-on workshops, science fairs, and public talks.

REFERENCES

Prof. Dr. Ejaz Ahmed,

Professor, School of Chemistry, University of The Punjab, Lahore, Pakistan

ejaz.ahmed.chem@pu.edu.pk

Prof. Dr. Muhammad Arshad,

Professor, Institute of Chemistry, The Islamia University of Bahawalpur, Pakistan

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